KARTHIK KRISHNA K U

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PROFESSIONAL SUMMARY

AI Engineer with hands-on experience building end-to-end Retrieval-Augmented Generation (RAG) systems using Large Language Models (LLMs). Skilled in designing production-ready, multi-tenant AI platforms with FastAPI, PyTorch, and Qdrant. Demonstrated ability to enhance model inference efficiency by 30% and deploy containerized AI microservices on Microsoft Azure for scalable, high-performance applications.

TECHNICAL SKILLS

Languages: Python, SQL
AI & Machine Learning: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), NLP, Prompt Engineering, Semantic Search, Text Embeddings, Transformer Models
Frameworks & Libraries: FastAPI, PyTorch, LangChain, LangGraph, Django, HuggingFace, SentenceTransformers
Databases & Vectors: Qdrant, MongoDB, PostgreSQL
DevOps & Tools: Docker, Git, Microsoft Azure, NumPy, Pandas

PROFESSIONAL EXPERIENCE

AI Engineer Intern | Safe Software and Integrated Solutions Pvt. Ltd. *Sept 2025 - Present*

- Architected document understanding and retrieval pipelines using LangChain and LangGraph.
- Built multi-tenant RAG systems with custom embeddings for enterprise knowledge platforms.
- Boosted model inference throughput by 30% via backend optimizations and efficient pipeline design.
- Deployed AI microservices with FastAPI and Docker on Microsoft Azure.
- Developed a full-stack Conversational AI with semantic memory and intent detection using FastAPI, Django, and Qdrant.
- Built VerbaLink, a real-time Voice AI Assistant integrating Twilio and Claude for multi-lingual support.

Python Developer | 21 Century Infoways Pvt. Ltd. *Sept 2023 - Aug 2025*

- Designed and built REST APIs connecting front-end interfaces with backend business logic.
- Optimized database schemas and queries to improve overall system performance.
- Integrated cloud APIs and third-party services into scalable backend infrastructure.

PROJECTS

Hashsense AI - Multi-Tenant Enterprise RAG Platform

- Built a multi-tenant RAG platform using FastAPI and Qdrant for secure, scalable semantic search.
- Implemented document ingestion pipelines, vector embeddings, and role-based access control for enterprise knowledge systems.
- Integrated intent detection and chatbot workflows for intelligent, context-aware query handling.

ChatDocs AI - GPU-Accelerated RAG System

- Developed a document QA system using Qdrant and SBERT embeddings for high-accuracy semantic retrieval.
- Optimized GPU-accelerated inference with CUDA and PyTorch, significantly reducing response latency.
- Built vector search pipelines for efficient large-scale document retrieval.

AI Voice Call Assistant - Conversational Voice AI System

- Built an AI-powered voice assistant for automated customer interactions using STT and TTS models.
- Developed backend services to process voice input, generate AI responses, and return synthesized speech in real time.

EDUCATION

B.Sc. - Artificial Intelligence and Machine Learning *2020 - 2023*
AJK College of Arts and Science, Palakkad, Kerala

Higher Secondary Education (Science) *2018 - 2020*
GHSS Kizhakkencherry, Palakkad

CERTIFICATIONS

Python for Machine Learning | Convolutional Neural Networks | Statistics for Data Science | Python and Data Analysis

LANGUAGES

English | Malayalam | Tamil